



Magnetic Reconnection

Scientists from Columbia University publish paper on how to use black holes as an energy source. <http://dgit.in/feb21-17>



Interstellar Vacation

Axiom announces first private space crew paying \$55 million for a trip to the ISS. <http://dgit.in/feb21-18>



GALAXY EVOLUTION

PART II

How the Milky Way changed over time

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It was only around the 1920s that “spiral nebulae” were considered distinct from the Milky Way and scientists began to accept the idea of “island universes” or galaxies as we now call them. Walter Baade, a German astronomer working at Mount Wilson Observatory in the United States, took advantage of black-outs during WWII to make detailed observations of the Andromeda galaxy. Towards the outer edges of the galaxy, he noticed blue stars which he called population I, and towards the center of the galaxy, he noticed more red stars which he called population II. Astronomers observing these stars discovered

something fundamentally different about the two types of stars. The stars at the edges of the galaxies were bluer, younger and richer in metals, while the stars towards the centre of galaxies were redder, older and poor in metals. The stars in the centers of galaxies were formed earlier in the universe than the stars at the edges.

In the 1960s, three astronomers Olin Eggen, Donald Lynden-Bell and Allan Sandage came up with the ELS model, which explained galactic formation based on the observations of population I and II stars in the Milky Way. The population I stars, like the sun are in the outer edges of the galaxies, and follow orbits along the galactic plane. The population II stars, closer to the center of the galaxy, in what is known as the bulge, had elliptical orbits that cut across the plane. The ELS model was able to explain the behavior of both populations of stars. According to the ELS model, a protogalaxy or a nebulous cloud of gas and matter began to accumulate and collapse towards its center, forming the first generation of metal poor stars,

we now know as the population I stars. Because of a combination of factors including energy release from the contracting cloud, and its rotational motion, the matter in the Milky Way settled into a disc, with the metal from the first supernovas towards the middle of the galaxy seeding a new generation of population I stars. According to the ELS model, this entire process took place over a period of only 300 million years. However, a study of the metallicity of the stars in the globular clusters revealed a variation in ages too far to be accommodated by how fast the Milky Way gained its shape in the ELS model. In 1978, Leonard Searle and Robert Zinn proposed an improved model where different clouds fragments of gas that had already seen phases of star formation, and had globular clusters embedded within them, accumulated to form the Milky Way. This model holds up to observations made even later, and the existence of dwarf galaxies around the Milky Way adds credence to the theory. The Sagittarius Dwarf Galaxy for example repeatedly passes the galactic plane, losing some of its mass during every pass, and will eventually be consumed wholly by the Milky Way. Early in the formation the Milky Way would have swallowed up more galactic cloud fragments, and those that didn't became the Dwarf Galaxies in our stellar neighbourhood. Throughout their life, galaxies keep swallowing up smaller galaxies, and the Milky Way is in the process of swallowing up the Small and Large Magellanic clouds.

Researchers now look at how old or young the stars are, as well as how they are moving to figure out the history of the Milky Way. In 2016, researchers used data gathered from the Sloan Digital Sky Survey, investigated the ages of 130,000 stars in the Milky Way, as well as their positions within the galaxy, and showed conclusively for